

RETAILMART V3 • SQL

Aggregates & GROUP BY — Quick Quiz

Quiz 1: What does this return?

EASY

ANSWER: `COUNT(*)` = 104,987 (all shipment rows). `COUNT(delivered_date)` = 90,005 (COUNT of a column skips NULLs). The gap - 14,982 - is exactly the shipments not yet delivered. `COUNT(*)` vs `COUNT(col)` is the fastest data-quality probe in SQL.

Question



ANSWER

```
SELECT COUNT(*), COUNT(delivered_date)
FROM sales.shipments;
```

Quiz 2: Why does this crash?

MEDIUM

ANSWER: WHERE cannot use aggregate functions — GROUP BY hasn't run yet at the WHERE stage. Fix: move the condition to HAVING COUNT(*) > 100.

Question



ANSWER

```
SELECT store_id, COUNT(*)  
FROM sales.orders  
WHERE COUNT(*) > 100  
GROUP BY store_id;
```

Quiz 3: What is wrong here?

MEDIUM

ANSWER: `order_status` is in `SELECT` but NOT in `GROUP BY` and NOT inside an aggregate. The Golden Rule: every non-aggregated column in `SELECT` must be in `GROUP BY`. Fix: add `order_status` to `GROUP BY`, or remove it, or wrap in `STRING_AGG`.

Question



ANSWER

```
SELECT store_id, order_status, COUNT(*)  
FROM sales.orders  
GROUP BY store_id;
```

Quiz 4: WHERE or HAVING?

HARD

Where do these conditions go?

- a) 'Only count delivered orders' → WHERE order_status = 'Delivered' (row-level filter, before grouping)
- b) 'Only show stores with 500+ orders' → HAVING COUNT(*) >= 500 (group-level filter, after grouping)
- c) 'Only orders from 2025' → WHERE order_date >= '2025-01-01' (row-level filter)
- d) 'Only months with revenue > 10 lakh' → HAVING SUM(net_total) > 1000000 (aggregate filter)

Quiz 5: Predict the Result

CRAZY

ANSWER: total = 150000, stores = 200 (unique store_ids), revenue_per_store = total revenue divided by 200. NULLIF prevents divide-by-zero. COUNT(DISTINCT) counts unique values only. ROUND(..., 0) gives whole number.

Question

 ANSWER

```
SELECT
  COUNT(*) AS total,
  COUNT(DISTINCT store_id) AS stores,
  ROUND(SUM(net_total) /
    NULLIF(COUNT(DISTINCT store_id),
      0), 0) AS revenue_per_store
FROM sales.orders;
```

Quiz 6: What is the output difference?

HARD

ANSWER: Both return the SAME number! Query A filters rows first (WHERE). Query B counts matching rows with CASE (conditional aggregation). Same result, different technique. Query B is useful when you need multiple counts in one query.

Question

 ANSWER

```
-- Query A:  
SELECT COUNT(*) FROM sales.orders  
WHERE order_status = 'Delivered';
```

```
-- Query B:  
SELECT COUNT(CASE WHEN  
    order_status = 'Delivered'  
    THEN 1 END) FROM sales.orders;
```